



Sampling and reconstruction in reproducing kernel subspaces of mixed Lebesgue spaces

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Abstract

In this paper, we study the sampling and average sampling problems in a reproducing kernel subspace of mixed Lebesgue space. Let V be an image of $L^{p,q}(\mathbb{R}^{d+1})$ under idempotent integral operator defined by a kernel K satisfying certain decay and regularity conditions. Then, we prove that every f in V can be reconstructed uniquely and stably from its samples as well as from its average samples taken on a sufficiently small γ -dense set. Further, we derive iterative reconstruction algorithms for reconstruction of f in V from its samples and average samples. We also obtain the error estimates in iterative reconstruction algorithm from noisy samples and iterative noise.

Keywords Nonuniform sampling · Reproducing kernel subspaces · Shift-invariant spaces · Average sampling · Reconstruction algorithms · Integral operator · Mixed Lebesgue spaces

Mathematics Subject Classification 42C15 · 94A20 · 47B34 · 32A70

1 Introduction

The mixed Lebesgue spaces $L^{p,q}(\mathbb{R}^{d+1})$ were introduced by Benedek and Panzone in [5] which are natural generalization of Lebesgue spaces $L^p(\mathbb{R}^{d+1})$. This space consists of multi-variable measurable functions with a norm defined in terms of different iterative L^p norms of the function with each variable. Ward [32] obtained mixed Lebesgue versions of several classical results including some inequalities and sam-

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pling theorems. Mixed Lebesgue spaces play a significant role in the study of singular integrals, time-based partial differential equations, sampling, average sampling, and time-varying signals (see [10,11,14,17,22]). In this context, we explore the sampling and average sampling problems in a reproducing kernel subspace of $L^{p,q}(\mathbb{R}^{d+1})$.

In this paper, we consider the sampling and average sampling problems in a function space V , where V is an image of $L^{p,q}(\mathbb{R}^{d+1})$ under idempotent integral operator defined by a kernel which satisfies certain decay and regularity conditions.

Sampling problem: To find conditions on a sampling set $X = \{(x_j, y_k) : j \in \mathbb{Z}^d, k \in \mathbb{Z}\}$ such that every $f \in V$ can be recovered in a stable and unique way by using its sample values $\{f(x_j, y_k) : j \in \mathbb{Z}^d, k \in \mathbb{Z}\}$. Mathematically, we are interested to establish the following sampling inequality

$$A\|f\|_{L^{p,q}} \leq \left(\sum_{k \in \mathbb{Z}} \left(\sum_{j \in \mathbb{Z}^d} |f(x_j, y_k)|^p \right)^{\frac{q}{p}} \right)^{\frac{1}{q}} \leq B\|f\|_{L^{p,q}}$$

for every $f \in V$ and for some $0 < A \leq B < \infty$.

Average sampling problem: To find conditions on a sampling set $X = \{(x_j, y_k) : j \in \mathbb{Z}^d, k \in \mathbb{Z}\}$ and averaging functions $\{\psi_{x_j, y_k} : j \in \mathbb{Z}^d, k \in \mathbb{Z}\}$ such that every $f \in V$ can be recovered in a stable and unique way by using its average sample values $\{\langle f, \psi_{x_j, y_k} \rangle : j \in \mathbb{Z}^d, k \in \mathbb{Z}\}$. Mathematically, we are interested to establish the following average sampling inequality

$$A\|f\|_{L^{p,q}} \leq \left(\sum_{k \in \mathbb{Z}} \left(\sum_{j \in \mathbb{Z}^d} |\langle f, \psi_{x_j, y_k} \rangle|^p \right)^{\frac{q}{p}} \right)^{\frac{1}{q}} \leq B\|f\|_{L^{p,q}}$$

for every $f \in V$ and for some $0 < A \leq B < \infty$. For a detailed study of the sampling problem, we refer to the survey articles by Butzer and Stens [8], Aldroubi and Gröchenig [2].

Sampling and reconstruction play a significant role in several research areas such as signal and modern digital data processing. The celebrated Shannon sampling theorem states that every $f \in L^2(\mathbb{R})$ such that $\text{supp}(\widehat{f}) \subset [-\Omega, \Omega]$, where

$$\widehat{f}(\xi) = \int_{\mathbb{R}} f(t) e^{-2\pi i t \xi} dt, \quad \xi \in \mathbb{R},$$

can be reconstructed stably and uniquely from its samples $\{f(\frac{k}{2\Omega}) : k \in \mathbb{Z}\}$ by using the formula

$$f(x) = \sum_{k \in \mathbb{Z}} f\left(\frac{k}{2\Omega}\right) \frac{\sin \pi(2\Omega x - k)}{\pi(2\Omega x - k)}, \quad x \in \mathbb{R}.$$

This theorem gives an explicit reconstruction formula, but due to slow decay and infinite support of the *sinc* function, $\text{sinc}(x) = \frac{\sin \pi x}{\pi x}$, it is often less suitable for

numerical implementation. Walter and Shen [31] obtained a sampling series expansion of a band-limited function by using the prolate spheroidal wave functions. An advantage of the prolate spheroidal wave functions over the *sinc* function is that they are more concentrated on finite intervals. Mounni and Zayed [18] obtained a generalization of the prolate spheroidal wave functions, and using this generalization, they established a sampling formula for a general class of band-limited functions. Nonuniform sampling sets are known in many practical situations such as in communication theory, medical imaging, and astronomical measurements etc. Razafinjato [21] studied the nonuniform sampling problem and proved that every $f \in L^2(\mathbb{R})$ such that $\text{supp}(\hat{f}) \subset [-\Omega, \Omega]$ can be reconstructed iteratively by using nonuniform samples of f and its derivatives. Sampling problem in the space of band-limited functions is also studied under generalized Fourier transform such as linear canonical transform and fractional Fourier transform (see [28,33,36]).

Sampling problem is extended to the shift-invariant spaces $V(\varphi) = \overline{\text{span}}\{\varphi(\cdot - k) : k \in \mathbb{Z}\}$. Walter [30] extended the Shannon sampling theorem for $V(\varphi)$ under mild assumptions on φ . Aldroubi and Gröchenig [1] studied the nonuniform sampling problem in $V(Q_m)$, where Q_m is a B -spline function of order m , and proved that every separated subset $X = \{x_j : j \in \mathbb{Z}\}$ of $\mathbb{R}$, such that $\sup_j (x_{j+1} - x_j) < 1$, is a stable set of sampling for $V(Q_m)$. Further, Xian et al. [35] gave an iterative approximation-projection algorithm for reconstruction of $f \in V(Q_m)$ from its nonuniform samples $\{f(x_j) : j \in \mathbb{Z}\}$. Both uniform and nonuniform sampling sets in general shift-invariant spaces are obtained by Kulkarni et al. in [15]. Recently, Selvan and Radha [24] obtained nonuniform sampling sets in shift-invariant spaces generated by continuously differentiable functions. In particular, they proved that if X is a separated subset of $\mathbb{R}$ such that $\sup_j (x_{j+1} - x_j) < 1$, then X is a stable set of sampling for $V(\phi)$, where ϕ is a Meyer scaling function. Sampling problem is also extended to the quasi shift-invariant spaces $V_X(\varphi) = \overline{\text{span}}\{\varphi(\cdot - x_k) : k \in \mathbb{Z}\}$, where $X = \{x_k : k \in \mathbb{Z}\}$ is any strictly increasing sequence of $\mathbb{R}$. Gröchenig and Stöckler [13] studied the sampling problem in quasi shift-invariant spaces generated by totally positive functions φ of finite type $M \geq 2$, i.e., $\hat{\varphi}(\xi) = \prod_{v=1}^M (1 + 2\pi i \delta_v \xi)^{-1}$ for $0 \neq \delta_v \in \mathbb{R}$, and $M \geq 2$. In particular, they proved that if $X = \alpha\mathbb{Z}$, $\alpha > 0$, then every separated subset $Y = \{y_k : k \in \mathbb{Z}\}$ of $\mathbb{R}$, such that $\sup_k (y_{k+1} - y_k) < \alpha$, is a stable set of sampling for $V_X(\varphi)$. Further, Gröchenig et al. [12] studied the nonuniform sampling problem in quasi shift-invariant spaces generated by totally positive functions φ of Gaussian type, i.e., $\hat{\varphi}(\xi) = \prod_{v=1}^M (1 + 2\pi i \delta_v \xi)^{-1} e^{-c\xi^2}$ for $\delta_v \in \mathbb{R}$, and $c > 0$. Recently, Cheng et al. [9] studied the distributed sampling and reconstruction for spatial signals. For a detailed study of the sampling problem in shift-invariant spaces associated with different transforms, we refer [2,3,6,7,23] and references therein.

Nashed and Sun [19] extended the sampling problem in a more general reproducing kernel subspace V_p of $L^p(\mathbb{R}^d)$. Here the reproducing kernel subspace V_p is the range space of an idempotent integral operator defined by a kernel K satisfying certain decay and regularity conditions. It is shown [19] that space V_p includes the shift-invariant spaces, quasi shift-invariant spaces, and the space of functions with finite rate of innovation. Vetterli et al. [29] studied the sampling problem in the space of signals with finite rate of innovation. Recently, Li et al. [17] studied the nonuniform

sampling and reconstruction in shift-invariant spaces of $L^{p,q}(\mathbb{R}^{d+1})$. In this paper, we consider the function space as a reproducing kernel subspace V of $L^{p,q}(\mathbb{R}^{d+1})$ of the form

$$V = \left\{ Tf(x, y) = \int_{\mathbb{R}} \int_{\mathbb{R}^d} K(x, y, r, s) f(r, s) dr ds : f \in L^{p,q}(\mathbb{R}^{d+1}), T^2 = T \right\}.$$

The reproducing kernel subspace V includes the space V_p [19], shift-invariant space $V_{p,q}(\varphi)$ [17]. Therefore, we address the sampling problem in a large space with flexible norm. In particular, we study the nonuniform sampling and reconstruction for functions in V .

In many practical situations, measurements are not the exact values of a function f on the sampling set X but are the local averages of f near (x_j, y_k) . Specifically, measurements can be taken as follows

$$\langle f, \psi_{x_j, y_k} \rangle = \int_{\mathbb{R}} \int_{\mathbb{R}^d} f(x, y) \overline{\psi_{x_j, y_k}(x, y)} dx dy,$$

where $\{\psi_{x_j, y_k} : j \in \mathbb{Z}^d, k \in \mathbb{Z}\}$ is a collection of averaging functions. By assuming the following conditions on ψ_{x_j, y_k} ,

$$(i) \text{supp}(\psi_{x_j, y_k}) \subset B_\delta(x_j, y_k), (ii) \int_{\mathbb{R}^{d+1}} \psi_{x_j, y_k}(x) dx = 1, \text{ and} \\ (iii) \int_{\mathbb{R}^{d+1}} |\psi_{x_j, y_k}(x)| dx \leq M < \infty,$$

one can observe the good approximation of the original function can be obtained from local averages for smaller values of δ . Sun and Zhou studied the reconstruction of band-limited functions and B -spline functions from local averages in [26,27] respectively. Aldroubi et al. [4] studied the convolution sampling and reconstruction problem in shift-invariant spaces. Sun [25] studied the nonuniform average sampling problem in the space of functions with finite rate of innovation. In [16], the authors studied average sampling and reconstruction in shift-invariant spaces. Xian [34] studied the average sampling problem in a reproducing kernel subspace V_p with respect to certain weights. Nashed et al. [20] studied the convolution sampling and reconstruction in a reproducing kernel subspaces V_p . Recently, Zhang [37] discussed the nonuniform average sampling problem in shift-invariant spaces of mixed Lebesgue spaces. In this paper, we discuss the average sampling problem in the reproducing kernel subspace V .

The paper is organized as follows. In Sect. 2, we mention the notations and required preliminary results for the paper. In Sect. 3, we discuss a verifiable condition for the boundedness of the integral operator on $L^{p,q}(\mathbb{R}^{d+1})$. The main results are presented in Sects. 4 and 5. First, we study the ideal sampling and reconstruction in V . Then, we study the average sampling and reconstruction problem in V . Moreover, we provide iterative reconstruction algorithms for the recovery of f in V from its samples

and average samples. Section 6 contains the error analysis in iterative reconstruction algorithm.

2 Notations and preliminaries

In this section, we give some definitions, notations and preliminary results required for the paper.

Definition 2.1 Let $1 \leq p, q < \infty$. The space of all measurable functions $f = f(x, y)$ defined on $\mathbb{R}^d \times \mathbb{R}$ such that

$$\|f\|_{L^{p,q}}^q = \int_{\mathbb{R}} \left(\int_{\mathbb{R}^d} |f(x, y)|^p dx \right)^{\frac{q}{p}} dy < \infty$$

is called the mixed Lebesgue space and denoted by $L^{p,q} = L^{p,q}(\mathbb{R}^{d+1})$.

For $1 \leq p, q < \infty$, the mixed sequence spaces $\ell^{p,q}$ are defined as

$$\ell^{p,q} = \ell^{p,q}(\mathbb{Z}^{d+1}) = \left\{ c = \{c(k_1, k_2)\} : \|c\|_{\ell^{p,q}}^q = \sum_{k_2 \in \mathbb{Z}} \left(\sum_{k_1 \in \mathbb{Z}^d} |c(k_1, k_2)|^p \right)^{\frac{q}{p}} < \infty \right\}.$$

It is easy to see that $L^{p,p}(\mathbb{R}^{d+1}) = L^p(\mathbb{R}^{d+1})$ and $\ell^{p,p}(\mathbb{Z}^{d+1}) = \ell^p(\mathbb{Z}^{d+1})$.

Definition 2.2 The mixed Wiener amalgam space $W(L^{p,q})$ consists of all measurable functions $f = f(x, y)$ such that

$$\|f\|_{W(L^{p,q})}^q = \sum_{n \in \mathbb{Z}} \sup_{y \in [0, 1]} \left(\sum_{k \in \mathbb{Z}^d} \sup_{x \in [0, 1]^d} |f(x + k, y + n)|^p \right)^{\frac{q}{p}} < \infty.$$

The classical Wiener amalgam space $W(L^p) = W(L^p)(\mathbb{R}^{d+1})$ consists of all measurable functions f such that

$$\|f\|_{W(L^p)}^p = \sum_{k \in \mathbb{Z}^{d+1}} \text{ess sup}_{x \in [0, 1]^{d+1}} |f(x + k)|^p < \infty.$$

It is easy to observe that $W(L^p) \subset W(L^{p,p})$. We denote the class of all continuous functions in $W(L^p)$ and $W(L^{p,q})$ by $W(C, L^p)$ and $W(C, L^{p,q})$ respectively.

Proposition 2.1 [5] Let $1 \leq p, q \leq \infty$. If $f \in L^{p,q}$ and $g \in L^{p',q'}$ with $\frac{1}{p} + \frac{1}{p'} = 1$ and $\frac{1}{q} + \frac{1}{q'} = 1$, then $fg \in L^{1,1}$. Moreover

$$\|fg\|_{L^{1,1}} \leq \|f\|_{L^{p,q}} \|g\|_{L^{p',q'}}.$$

Proposition 2.2 [17] *If $\varphi \in W(L^{1,1})$ satisfies*

$$\sum_{k \in \mathbb{Z}^{d+1}} |\widehat{\varphi}(\xi + 2\pi k)|^2 > 0, \quad \xi \in \mathbb{R}^{d+1},$$

then there exist a function $\widetilde{\varphi}$

$$\widetilde{\varphi} = \sum_{k_2 \in \mathbb{Z}} \sum_{k_1 \in \mathbb{Z}^d} d(k_1, k_2) \varphi(\cdot - k_1, \cdot - k_2)$$

such that

$$\int_{\mathbb{R}} \int_{\mathbb{R}^d} \varphi(r - k_1, s - k_2) \overline{\widetilde{\varphi}(r - l_1, s - l_2)} dr ds = \delta_{(k_1, k_2), (l_1, l_2)},$$

where $d = \{d(k_1, k_2) : k_1 \in \mathbb{Z}^d, k_2 \in \mathbb{Z}\} \in \ell^1$, and $\delta_{(k_1, k_2), (l_1, l_2)}$ is the Kronecker delta function. Moreover, $\widetilde{\varphi}$ belongs to $W(L^{1,1})$. In addition, if $\varphi \in W(L^1)$, then $\widetilde{\varphi} \in W(L^1)$.

Definition 2.3 For $\gamma > 0$, modulus of continuity (or γ -oscillation) of a function f is defined as

$$\text{osc}_\gamma f(x) = \sup_{x' \in \overline{B_\gamma(0)}} |f(x + x') - f(x)|,$$

where $B_\gamma(0)$ denotes the ball centered at zero and radius γ .

Lemma 2.1 [2] *Let $\varphi \in W(C, L^1)$, then $\text{osc}_\gamma \varphi \in W(C, L^1)$.*

In the following proposition, we present an iterative reconstruction algorithm with exponential convergence using Neumann series expansion.

Proposition 2.3 *Let S be a bounded operator on a Banach space $\mathcal{M}$ such that*

$$\|f - Sf\|_{\mathcal{M}} \leq r \|f\|_{\mathcal{M}}$$

holds for every $f \in \mathcal{M}$ and for some $r \in (0, 1)$. Then S is invertible on $\mathcal{M}$ and $S^{-1} = \sum_{k=0}^{\infty} (I - S)^k$. Moreover, f can be reconstructed by using the following iterative reconstruction algorithm

$$\begin{aligned} f_0 &= Sf, \\ f_{n+1} &= f_n + S(f - f_n), \quad n \geq 0. \end{aligned}$$

Then we have $f_n \rightarrow f$ and the error estimate after n^{th} iteration is given by

$$\|f - f_n\|_{\mathcal{M}} \leq r^{n+1} \|f\|_{\mathcal{M}}.$$

3 Reproducing kernel subspaces of mixed Lebesgue spaces

Definition 3.1 A closed subspace V of $L^{p,q}$ is called a reproducing kernel subspace if the point-wise evaluation functionals on V are continuous, i.e., for any $(x, y) \in \mathbb{R}^d \times \mathbb{R}$ there exists a constant $C_{x,y} > 0$ such that

$$|f(x, y)| \leq C_{x,y} \|f\|_{L^{p,q}}, \quad \forall f \in V.$$

Let $1 \leq p \leq q < \infty$ and $T : L^{p,q} \rightarrow L^{p,q}$ be an idempotent integral operator defined by

$$Tf(x, y) = \int_{\mathbb{R}} \int_{\mathbb{R}^d} K(x, y, r, s) f(r, s) dr ds, \quad (3.1)$$

where K is a measurable kernel such that

$$\begin{aligned} \|K\|_{op} &= \sup_y \sup_x \|K(x, y, \cdot, \cdot)\|_{L_{r,s}^{1,1}}^{1-\frac{1}{p}} \times \sup_y \left\| \sup_r \|K(\cdot, y, r, \cdot)\|_{L_x^1} \right\|_{L_s^{\frac{q-p}{pq}}}^{q-p} \\ &\times \sup_s \left\| \sup_r \|K(\cdot, \cdot, r, s)\|_{L_x^1} \right\|_{L_y^1}^{\frac{1}{q}} < \infty. \end{aligned} \quad (3.2)$$

Now we establish a verifiable condition for the boundedness of the integral operator defined by (3.1) by making use of Hölder's inequality. In particular, we prove the following result.

Theorem 3.1 Let $1 \leq p \leq q < \infty$ and K be a measurable kernel satisfying the decay condition (3.2). Then the linear operator $T : L^{p,q} \rightarrow L^{p,q}$ defined by

$$Tf(x, y) = \int_{\mathbb{R}} \int_{\mathbb{R}^d} K(x, y, r, s) f(r, s) dr ds$$

is a bounded operator and $\|T\|_{L^{p,q}} \leq \|K\|_{op}$.

Proof Since

$$\begin{aligned} |Tf(x, y)| &\leq \int_{\mathbb{R}} \int_{\mathbb{R}^d} |K(x, y, r, s)| |f(r, s)| dr ds \\ &\leq \int_{\mathbb{R}} \left(\int_{\mathbb{R}^d} |K(x, y, r, s)| dr \right)^{\frac{1}{p'}} \left(\int_{\mathbb{R}^d} |K(x, y, r, s)| |f(r, s)|^p dr \right)^{\frac{1}{p}} ds \\ &\leq \left(\int_{\mathbb{R}} \int_{\mathbb{R}^d} |K(x, y, r, s)| dr ds \right)^{\frac{1}{p'}} \left(\int_{\mathbb{R}} \int_{\mathbb{R}^d} |K(x, y, r, s)| |f(r, s)|^p dr ds \right)^{\frac{1}{p}}, \end{aligned}$$

where $\frac{1}{p} + \frac{1}{p'} = 1$, we have

$$\begin{aligned}
 & \int_{\mathbb{R}^d} |Tf(x, y)|^p dx \\
 & \leq \int_{\mathbb{R}^d} \left(\int_{\mathbb{R}} \int_{\mathbb{R}^d} |K(x, y, r, s)| dr ds \right)^{\frac{p}{p'}} \left(\int_{\mathbb{R}} \int_{\mathbb{R}^d} |K(x, y, r, s)| |f(r, s)|^p dr ds \right) dx \\
 & \leq \sup_x \|K(x, y, \cdot, \cdot)\|_{L_{r,s}^{1,1}}^{\frac{p}{p'}} \int_{\mathbb{R}} \int_{\mathbb{R}^d} \|K(\cdot, y, r, s)\|_{L_x^1} |f(r, s)|^p dr ds \\
 & \leq \sup_x \|K(x, y, \cdot, \cdot)\|_{L_{r,s}^{1,1}}^{\frac{p}{p'}} \int_{\mathbb{R}} \sup_r \|K(\cdot, y, r, s)\|_{L_x^1} \int_{\mathbb{R}^d} |f(r, s)|^p dr ds \\
 & \leq \sup_x \|K(x, y, \cdot, \cdot)\|_{L_{r,s}^{1,1}}^{\frac{p}{p'}} \left(\int_{\mathbb{R}} \sup_r \|K(\cdot, y, r, s)\|_{L_x^1} ds \right)^{\frac{q-p}{q}} \\
 & \quad \times \left(\int_{\mathbb{R}} \sup_r \|K(\cdot, y, r, s)\|_{L_x^1} \left(\int_{\mathbb{R}^d} |f(r, s)|^p dr \right)^{\frac{q}{p}} ds \right)^{\frac{p}{q}}.
 \end{aligned}$$

Hence,

$$\begin{aligned}
 \left(\int_{\mathbb{R}^d} |Tf(x, y)|^p dx \right)^{\frac{q}{p}} & \leq \sup_x \|K(x, y, \cdot, \cdot)\|_{L_{r,s}^{1,1}}^{\frac{q}{p'}} \times \left\| \sup_r \|K(\cdot, y, r, \cdot)\|_{L_x^1} \right\|_{L_s^1}^{\frac{q-p}{p}} \\
 & \quad \times \int_{\mathbb{R}} \sup_r \|K(\cdot, y, r, s)\|_{L_x^1} \left(\int_{\mathbb{R}^d} |f(r, s)|^p dr \right)^{\frac{q}{p}} ds.
 \end{aligned}$$

Consider

$$\begin{aligned}
 \|Tf\|_{L^{p,q}}^q & = \int_{\mathbb{R}} \left(\int_{\mathbb{R}^d} |Tf(x, y)|^p dx \right)^{\frac{q}{p}} dy \\
 & \leq \sup_y \sup_x \|K(x, y, \cdot, \cdot)\|_{L_{r,s}^{1,1}}^{\frac{q}{p'}} \times \sup_y \left\| \sup_r \|K(\cdot, y, r, \cdot)\|_{L_x^1} \right\|_{L_s^1}^{\frac{q-p}{p}} \\
 & \quad \times \sup_s \left\| \sup_r \|K(\cdot, \cdot, r, s)\|_{L_x^1} \right\|_{L_y^1} \|f\|_{L^{p,q}}^q.
 \end{aligned}$$

Thus, we obtain

$$\begin{aligned}
 \|Tf\|_{L^{p,q}} & \leq \sup_y \sup_x \|K(x, y, \cdot, \cdot)\|_{L_{r,s}^{1,1}}^{\frac{1}{p'}} \times \sup_y \left\| \sup_r \|K(\cdot, y, r, \cdot)\|_{L_x^1} \right\|_{L_s^1}^{\frac{q-p}{pq}} \\
 & \quad \times \sup_s \left\| \sup_r \|K(\cdot, \cdot, r, s)\|_{L_x^1} \right\|_{L_y^1}^{\frac{1}{q}} \|f\|_{L^{p,q}} \\
 & = \|K\|_{op} \|f\|_{L^{p,q}},
 \end{aligned}$$

which completes the proof. $\square$

- Remark 3.1** (i) Samarah et al. [22] obtained the boundedness of the integral operator defined by a kernel K satisfying certain conditions involving the existence of dominating functions, but these conditions are not easily verifiable.
- (ii) Jiang and Sun [14] obtained the boundedness condition of the integral operator for $1 \leq p, q < \infty$. Our result is for $1 \leq p \leq q < \infty$, but we obtained a better bound than in [14]. For example, consider the kernel defined by

$$K(x, y, r, s) = \begin{cases} e^{-(y^2+r^2)} \frac{1}{\sqrt{|x-s|}} \chi_{[-1,1]}(s), & 0 < |x-s| \leq 1, \\ 0, & \text{otherwise,} \end{cases}$$

then K satisfies the condition given in (3.2) for $1 \leq p \leq q < \infty$, whereas K does not satisfy the condition given in [14, equation (2.1)]. At the same time, if the kernel K satisfies the condition in [14, equation (2.1)] then it also satisfies our proposed condition for boundedness. Thus, our class of kernels is bigger than the class of kernels considered in [14] for $1 \leq p \leq q < \infty$.

Consider the range space of T defined as

$$V = \{Tf : f \in L^{p,q}\}.$$

We also assume

$$\|K(x, y, \cdot, \cdot)\|_{L^{p',q'}} < \infty, \quad \forall x \in \mathbb{R}^d, y \in \mathbb{R}, \quad (3.3)$$

where $\frac{1}{p} + \frac{1}{p'} = 1$ and $\frac{1}{q} + \frac{1}{q'} = 1$. In addition,

$$\lim_{\gamma \rightarrow 0} \|\text{OSC}_\gamma K\|_{op} = 0, \quad (3.4)$$

where $\text{OSC}_\gamma K(x, y, r, s) = \sup_{(x', y') \in B_\gamma(0,0)} |K(x + x', y + y', r, s) - K(x, y, r, s)|$.

Theorem 3.2 *If an integral kernel K satisfies (3.2) and (3.3), then the range space V is a reproducing kernel subspace of $L^{p,q}$.*

Example 3.1 Let $\varphi \in W(L^{1,1})$ and $\tilde{\varphi} \in W(L^{1,1})$ be the dual of φ such that

$$\int_{\mathbb{R}} \int_{\mathbb{R}^d} \varphi(r - k_1, s - k_2) \overline{\tilde{\varphi}(r - l_1, s - l_2)} dr ds = \delta_{(k_1, k_2), (l_1, l_2)}.$$

For $p < q$, consider the shift-invariant space in mixed Lebesgue space $L^{p,q}$ defined in [17] as

$$V_{p,q}(\varphi) = \left\{ \sum_{k_2 \in \mathbb{Z}} \sum_{k_1 \in \mathbb{Z}^d} c(k_1, k_2) \varphi(\cdot - k_1, \cdot - k_2) : \{c(k_1, k_2) : k_1 \in \mathbb{Z}^d, k_2 \in \mathbb{Z}\} \in \ell^{p,q} \right\}.$$

Define the integral operator

$$T_\varphi f(x, y) = \int_{\mathbb{R}} \int_{\mathbb{R}^d} K_\varphi(x, y, r, s) f(r, s) dr ds,$$

where

$$K_\varphi(x, y, r, s) = \sum_{k_2 \in \mathbb{Z}} \sum_{k_1 \in \mathbb{Z}^d} \varphi(x - k_1, y - k_2) \overline{\widetilde{\varphi}(r - k_1, s - k_2)}.$$

Then $V_\varphi = \{T_\varphi f : f \in L^{p,q}\}$ is identical with $V_{p,q}(\varphi)$. Moreover, $V_{p,q}(\varphi)$ is a reproducing kernel subspace of $L^{p,q}$ ($p < q$).

Indeed, for $f \in V_{p,q}(\varphi)$, there exists $\{c(k_1, k_2)\} \in \ell^{p,q}$ such that

$$f(x, y) = \sum_{k_2 \in \mathbb{Z}} \sum_{k_1 \in \mathbb{Z}^d} c(k_1, k_2) \varphi(x - k_1, y - k_2).$$

Thus, for every $f \in V_{p,q}(\varphi)$, we have

$$\begin{aligned} T_\varphi f(x, y) &= \int_{\mathbb{R}} \int_{\mathbb{R}^d} K_\varphi(x, y, r, s) f(r, s) dr ds \\ &= \int_{\mathbb{R}} \int_{\mathbb{R}^d} \left(\sum_{l_2 \in \mathbb{Z}} \sum_{l_1 \in \mathbb{Z}^d} \varphi(x - l_1, y - l_2) \overline{\widetilde{\varphi}(r - l_1, s - l_2)} \right) \\ &\quad \times \left(\sum_{k_2 \in \mathbb{Z}} \sum_{k_1 \in \mathbb{Z}^d} c(k_1, k_2) \varphi(r - k_1, s - k_2) \right) dr ds \\ &= \sum_{l_2 \in \mathbb{Z}} \sum_{l_1 \in \mathbb{Z}^d} \varphi(x - l_1, y - l_2) \sum_{k_2 \in \mathbb{Z}} \sum_{k_1 \in \mathbb{Z}^d} c(k_1, k_2) \\ &\quad \int_{\mathbb{R}} \int_{\mathbb{R}^d} \varphi(r - k_1, s - k_2) \overline{\widetilde{\varphi}(r - l_1, s - l_2)} dr ds \\ &= \sum_{l_2 \in \mathbb{Z}} \sum_{l_1 \in \mathbb{Z}^d} c(l_1, l_2) \varphi(x - l_1, y - l_2) = f(x, y). \end{aligned}$$

Since $\varphi \in W(L^{1,1})$, we have [17] for any $c \in \ell^{p,q}$, the function defined by

$$f(x, y) = \sum_{k_2 \in \mathbb{Z}} \sum_{k_1 \in \mathbb{Z}^d} c(k_1, k_2) \varphi(x - k_1, y - k_2)$$

belongs to $L^{p,q}$. Moreover,

$$\|f\|_{L^{p,q}} \leq \|c\|_{\ell^{p,q}} \|\varphi\|_{W(L^{1,1})}.$$

This shows

$$V_{p,q}(\varphi) \subseteq V_\varphi. \quad (3.5)$$

Now for any $f \in V_\varphi$, we have

$$f(x, y) = \sum_{k_2 \in \mathbb{Z}} \sum_{k_1 \in \mathbb{Z}^d} \varphi(x - k_1, y - k_2) \int_{\mathbb{R}} \int_{\mathbb{R}^d} f(r, s) \overline{\widetilde{\varphi}(r - k_1, s - k_2)} dr ds.$$

Define the sequence

$$d(k_1, k_2) = \int_{\mathbb{R}} \int_{\mathbb{R}^d} f(r, s) \overline{\widetilde{\varphi}(r - k_1, s - k_2)} dr ds.$$

Then, from [17]

$$\|d\|_{\ell^{p,q}} \leq \|f\|_{L^{p,q}} \|\widetilde{\varphi}\|_{W(L^{1,1})}.$$

Thus, we obtain

$$f(x, y) = \sum_{k_2 \in \mathbb{Z}} \sum_{k_1 \in \mathbb{Z}^d} d(k_1, k_2) \varphi(x - k_1, y - k_2).$$

Hence

$$V_\varphi \subseteq V_{p,q}(\varphi). \quad (3.6)$$

From (3.5) and (3.6), we have $V_\varphi \equiv V_{p,q}(\varphi)$. Now, for any $f \in V_\varphi$,

$$\begin{aligned} |f(x, y)| &\leq \sum_{k_2 \in \mathbb{Z}} \sum_{k_1 \in \mathbb{Z}^d} |d(k_1, k_2)| |\varphi(x - k_1, y - k_2)| \\ &\leq \sum_{k_2 \in \mathbb{Z}} \left(\sum_{k_1 \in \mathbb{Z}^d} |d(k_1, k_2)|^p |\varphi(x - k_1, y - k_2)| \right)^{\frac{1}{p}} \\ &\quad \left(\sum_{k_1 \in \mathbb{Z}^d} |\varphi(x - k_1, y - k_2)| \right)^{1 - \frac{1}{p}} \\ &\leq \left(\sum_{k_2 \in \mathbb{Z}} \sum_{k_1 \in \mathbb{Z}^d} |d(k_1, k_2)|^p |\varphi(x - k_1, y - k_2)| \right)^{\frac{1}{p}} \|\varphi\|_{W(L^{1,1})}^{1 - \frac{1}{p}} \\ &\leq \left(\sum_{k_2 \in \mathbb{Z}} \sup_{k_1} |\varphi(x - k_1, y - k_2)| \sum_{k_1 \in \mathbb{Z}^d} |d(k_1, k_2)|^p \right)^{\frac{1}{p}} \|\varphi\|_{W(L^{1,1})}^{1 - \frac{1}{p}} \end{aligned}$$

$$\leq \left(\sum_{k_2 \in \mathbb{Z}} \sum_{k_1 \in \mathbb{Z}^d} |\varphi(x - k_1, y - k_2)|^{\frac{q}{q-p}} \right)^{\frac{q-p}{pq}} \|d\|_{\ell^{p,q}} \|\varphi\|_{W(L^{1,1})}^{1-\frac{1}{p}}.$$

Since $\varphi \in W(L^{1,1})$, for every $(x, y) \in \mathbb{R}^d \times \mathbb{R}$,

$$\sum_{k_2 \in \mathbb{Z}} \sum_{k_1 \in \mathbb{Z}^d} |\varphi(x - k_1, y - k_2)| \leq \|\varphi\|_{W(L^{1,1})} < \infty.$$

Therefore, for every $(x, y) \in \mathbb{R}^d \times \mathbb{R}$, $\exists N_{x,y} > 0$ such that

$$\sum_{|k_2| \geq N_{x,y}} \sum_{|k_1| \geq N_{x,y}} |\varphi(x - k_1, y - k_2)| < 1.$$

Thus,

$$\sum_{k_2 \in \mathbb{Z}} \sum_{k_1 \in \mathbb{Z}^d} |\varphi(x - k_1, y - k_2)|^{\frac{q}{q-p}} < M_{x,y} + 1,$$

where $M_{x,y} = \sum_{|k_2| < N_{x,y}} \sum_{|k_1| < N_{x,y}} |\varphi(x - k_1, y - k_2)|^{\frac{q}{q-p}}$. Hence, for every $f \in V_\varphi$,

$$|f(x, y)| \leq (M_{x,y} + 1)^{\frac{q-p}{pq}} \|\tilde{\varphi}\|_{W(L^{1,1})} \|\varphi\|_{W(L^{1,1})}^{1-\frac{1}{p}} \|f\|_{L^{p,q}},$$

which shows $V_{p,q}(\varphi)$ is a reproducing kernel subspace of $L^{p,q}$ ($p < q$).

4 Ideal sampling and reconstruction in V

In this section, we study the ideal sampling and reconstruction in general reproducing kernel subspace of mixed Lebesgue space.

Definition 4.1 A subset $\{(x_j, y_k) : j \in \mathbb{Z}^d, k \in \mathbb{Z}\}$ of $\mathbb{R}^{d+1}$ is called separated if $\inf_{(j,k) \neq (j',k')} \|(x_j, y_k) - (x_{j'}, y_{k'})\|_{\ell^2} = \eta > 0$.

Definition 4.2 A subset $\{(x_j, y_k) : j \in \mathbb{Z}^d, k \in \mathbb{Z}\}$ is said to be γ -dense in $\mathbb{R}^{d+1}$ if

$$\mathbb{R}^{d+1} = \cup_{j,k} B_r(x_j, y_k) \quad \text{for every } r \geq \gamma,$$

where $B_r(x_j, y_k)$ denotes the open ball centered at (x_j, y_k) with radius r .

A bounded partition of unity associated with open ball $\{B_\gamma(x_j, y_k) : j \in \mathbb{Z}^d, k \in \mathbb{Z}\}$ is a set of functions $\{\beta_{j,k} : j \in \mathbb{Z}^d, k \in \mathbb{Z}\}$ such that

- (i) $0 \leq \beta_{j,k} \leq 1, \forall j \in \mathbb{Z}^d, k \in \mathbb{Z}$,
- (ii) $\text{supp}(\beta_{j,k}) \subset B_\gamma(x_j, y_k)$,

$$(iii) \sum_{k \in \mathbb{Z}} \sum_{j \in \mathbb{Z}^d} \beta_{j,k} = 1.$$

In the following theorem, we establish the sampling inequality for functions belonging to the space V . For this, we first approximate the function f from its samples by using the quasi approximation Qf and then estimate the $\|f - Qf\|_{L^{p,q}}$ and $\|Qf\|_{L^{p,q}}$. By using these estimates we obtained the required lower bound in the sampling inequality. In particular, the theorem states as follows.

Theorem 4.1 *Let $1 \leq p \leq q < \infty$ and K be an integral kernel satisfying (3.2)–(3.4). If $X = \{(x_j, y_k) : j \in \mathbb{Z}^d, k \in \mathbb{Z}\}$ is a separated γ -dense set such that $\|OSC_\gamma K\|_{op} < 1$, then the following sampling inequality*

$$\begin{aligned} \left(\frac{1 - \|OSC_\gamma K\|_{op}}{C(p, q, \gamma, \eta, d)} \right) \|f\|_{L^{p,q}} &\leq \left(\sum_{k \in \mathbb{Z}} \left(\sum_{j \in \mathbb{Z}^d} |f(x_j, y_k)|^p \right)^{\frac{q}{p}} \right)^{\frac{1}{q}} \\ &\leq \frac{2^{1-\frac{1}{q}}}{D(p, q, \eta, d)} \left(1 + \|OSC_{\frac{\eta}{2}} K\|_{op}^q \right)^{\frac{1}{q}} \|f\|_{L^{p,q}} \end{aligned}$$

holds for every $f \in V$.

Proof For any $f \in V$, define the approximation operator

$$Qf = \sum_{k \in \mathbb{Z}} \sum_{j \in \mathbb{Z}^d} f(x_j, y_k) \beta_{j,k}.$$

Since

$$\begin{aligned} |f(x, y) - Qf(x, y)| &\leq \sum_{k \in \mathbb{Z}} \sum_{j \in \mathbb{Z}^d} |f(x, y) - f(x_j, y_k)| \beta_{j,k}(x, y) \\ &\leq \sum_{k \in \mathbb{Z}} \sum_{j \in \mathbb{Z}^d} \text{osc}_\gamma f(x, y) \beta_{j,k}(x, y) = \text{osc}_\gamma f(x, y), \end{aligned}$$

we have

$$\|f - Qf\|_{L^{p,q}}^q \leq \|\text{osc}_\gamma f\|_{L^{p,q}}^q.$$

Since

$$\text{osc}_\gamma f(x, y) \leq \int_{\mathbb{R}} \int_{\mathbb{R}^d} \text{OSC}_\gamma K(x, y, r, s) |f(r, s)| dr ds,$$

using the similar idea as in Theorem 3.1, we obtain

$$\begin{aligned}
 & \|f - Qf\|_{L^{p,q}} \\
 & \leq \sup_y \sup_x \|\text{OSC}_\gamma K(x, y, \cdot, \cdot)\|_{L_{r,s}^{1,1}}^{\frac{1}{p'}} \times \sup_y \sup_r \|\text{OSC}_\gamma K(\cdot, y, r, \cdot)\|_{L_x^1}^{\frac{q-p}{pq}} \\
 & \quad \times \sup_s \sup_r \|\text{OSC}_\gamma K(\cdot, \cdot, r, s)\|_{L_x^1}^{\frac{1}{q}} \|f\|_{L^{p,q}} \\
 & = \|\text{OSC}_\gamma K\|_{op} \|f\|_{L^{p,q}}.
 \end{aligned} \tag{4.1}$$

Consider

$$\begin{aligned}
 & \int_{\mathbb{R}^d} |Qf(x, y)|^p dx \\
 & = \int_{\mathbb{R}^d} \left| \sum_{k \in \mathbb{Z}} \sum_{j \in \mathbb{Z}^d} f(x_j, y_k) \beta_{j,k}(x, y) \right|^p dx \\
 & \leq \int_{\mathbb{R}^d} \left(\sum_{k \in \mathbb{Z}} \sum_{j \in \mathbb{Z}^d} |f(x_j, y_k) \beta_{j,k}(x, y)| \right)^p dx \\
 & \leq \int_{\mathbb{R}^d} \left(\sum_{k \in \mathbb{Z}} \sum_{j \in \mathbb{Z}^d} |f(x_j, y_k)|^p \beta_{j,k}(x, y) \right) \left(\sum_{k \in \mathbb{Z}} \sum_{j \in \mathbb{Z}^d} \beta_{j,k}(x, y) \right)^{\frac{p}{p'}} dx \\
 & \leq \sum_{k \in \mathbb{Z}} \sum_{j \in \mathbb{Z}^d} |f(x_j, y_k)|^p \|\beta_{j,k}(\cdot, y)\|_{L_x^1},
 \end{aligned}$$

where $\frac{1}{p} + \frac{1}{p'} = 1$. Thus, we have

$$\begin{aligned}
 \|Qf\|_{L^{p,q}}^q & = \int_{\mathbb{R}} \left(\int_{\mathbb{R}^d} |Qf(x, y)|^p dx \right)^{\frac{q}{p}} dy \\
 & \leq \int_{\mathbb{R}} \left(\sum_{k \in \mathbb{Z}} \sum_{j \in \mathbb{Z}^d} |f(x_j, y_k)|^p \|\beta_{j,k}(\cdot, y)\|_{L_x^1} \right)^{\frac{q}{p}} dy \\
 & \leq \int_{\mathbb{R}} \left[\sum_{k \in \mathbb{Z}} \left(\sum_{j \in \mathbb{Z}^d} |f(x_j, y_k)|^p \right) \sup_j \|\beta_{j,k}(\cdot, y)\|_{L_x^1} \right]^{\frac{q}{p}} dy \\
 & \leq \int_{\mathbb{R}} \left[\sum_{k \in \mathbb{Z}} \left(\sum_{j \in \mathbb{Z}^d} |f(x_j, y_k)|^p \right) \sup_j \|\beta_{j,k}(\cdot, y)\|_{L_x^1} \right]^{\frac{q}{p}} dy
 \end{aligned}$$

$$\begin{aligned}
& \left[\sum_{k \in \mathbb{Z}} \sup_j \|\beta_{j,k}(\cdot, y)\|_{L_x^1} \right]^{\frac{q-p}{p}} dy \\
& \leq \sup_y \left[\sum_{k \in \mathbb{Z}} \sup_j \|\beta_{j,k}(\cdot, y)\|_{L_x^1} \right]^{\frac{q-p}{p}} \\
& \quad \times \sum_{k \in \mathbb{Z}} \left[\left(\sum_{j \in \mathbb{Z}} |f(x_j, y_k)|^p \right)^{\frac{q}{p}} \int_{\mathbb{R}} \sup_j \|\beta_{j,k}(\cdot, y)\|_{L_x^1} dy \right] \\
& \leq \sup_y \left[\sum_{k \in \mathbb{Z}} \sup_j \|\beta_{j,k}(\cdot, y)\|_{L_x^1} \right]^{\frac{q-p}{p}} \sup_k \sup_j \|\beta_{j,k}\|_{L_x^1} \|L_y^1 \\
& \quad \sum_{k \in \mathbb{Z}} \left(\sum_{j \in \mathbb{Z}} |f(x_j, y_k)|^p \right)^{\frac{q}{p}}.
\end{aligned}$$

Hence, there exists a constant $C(p, q, \gamma, \eta, d)$ such that

$$\|Qf\|_{L^{p,q}} \leq C(p, q, \gamma, \eta, d) \left(\sum_{k \in \mathbb{Z}} \left(\sum_{j \in \mathbb{Z}^d} |f(x_j, y_k)|^p \right)^{\frac{q}{p}} \right)^{\frac{1}{q}}, \quad (4.2)$$

where $\sup_y \left[\sum_{k \in \mathbb{Z}} \sup_j \|\beta_{j,k}(\cdot, y)\|_{L_x^1} \right]^{\frac{q-p}{pq}} \sup_k \sup_j \|\beta_{j,k}\|_{L_x^1} \frac{1}{L_y^1} \leq C(p, q, \gamma, \eta, d)$.

Using (4.1) and (4.2), we have

$$\left(\sum_{k \in \mathbb{Z}} \left(\sum_{j \in \mathbb{Z}^d} |f(x_j, y_k)|^p \right)^{\frac{q}{p}} \right)^{\frac{1}{q}} \geq \left(\frac{1 - \|\text{OSC}_\gamma K\|_{op}}{C(p, q, \gamma, \eta, d)} \right) \|f\|_{L^{p,q}}, \quad (4.3)$$

for every $f \in V$.

Let $(x, y) \in B_{\frac{\eta}{2}}(x_j, y_k)$. For fix $y \in (y_k - \frac{\eta}{2}, y_k + \frac{\eta}{2})$, define $E_y = \{x : (x, y) \in B_{\frac{\eta}{2}}(x_j, y_k)\}$. Thus, we have

$$\begin{aligned}
\int_{E_y} |f(x_j, y_k)|^p dx & \leq 2^{p-1} \left[\int_{E_y} |f(x_j, y_k) - f(x, y)|^p dx + \int_{E_y} |f(x, y)|^p dx \right] \\
& \leq 2^{p-1} \left[\int_{E_y} |\text{osc}_{\frac{\eta}{2}} f(x, y)|^p dx + \int_{E_y} |f(x, y)|^p dx \right]
\end{aligned}$$

By taking summation over $j \in \mathbb{Z}^d$, we obtain

$$\sum_{j \in \mathbb{Z}^d} |f(x_j, y_k)|^p \int_{E_y} dx \leq 2^{p-1} \left[\int_{\mathbb{R}^d} |\operatorname{osc}_{\frac{\eta}{2}} f(x, y)|^p dx + \int_{\mathbb{R}^d} |f(x, y)|^p dx \right].$$

Thus, we have

$$\begin{aligned} & \left(\sum_{j \in \mathbb{Z}^d} |f(x_j, y_k)|^p \int_{E_y} dx \right)^{\frac{q}{p}} \\ & \leq 2^{q-1} \left[\left(\int_{\mathbb{R}^d} |\operatorname{osc}_{\frac{\eta}{2}} f(x, y)|^p dx \right)^{\frac{q}{p}} + \left(\int_{\mathbb{R}^d} |f(x, y)|^p dx \right)^{\frac{q}{p}} \right]. \end{aligned}$$

Hence

$$\begin{aligned} & \int_{y_k - \frac{\eta}{2}}^{y_k + \frac{\eta}{2}} \left(\sum_{j \in \mathbb{Z}} |f(x_j, y_k)|^p \int_{E_y} dx \right)^{\frac{q}{p}} dy \\ & \leq 2^{q-1} \left[\int_{y_k - \frac{\eta}{2}}^{y_k + \frac{\eta}{2}} \left(\int_{\mathbb{R}^d} |\operatorname{osc}_{\frac{\eta}{2}} f(x, y)|^p dx \right)^{\frac{q}{p}} dy \right. \\ & \quad \left. + \int_{y_k - \frac{\eta}{2}}^{y_k + \frac{\eta}{2}} \left(\int_{\mathbb{R}^d} |f(x, y)|^p dx \right)^{\frac{q}{p}} dy \right]. \end{aligned}$$

Now by taking summation over $k \in \mathbb{Z}$, we get

$$\begin{aligned} & D(p, q, \eta, d)^q \sum_{k \in \mathbb{Z}} \left(\sum_{j \in \mathbb{Z}^d} |f(x_j, y_k)|^p \right)^{\frac{q}{p}} \\ & \leq 2^{q-1} \left(\|\operatorname{osc}_{\frac{\eta}{2}} f\|_{L^{p,q}}^q + \|f\|_{L^{p,q}}^q \right) \\ & \leq 2^{q-1} \left(\|\operatorname{OSC}_{\frac{\eta}{2}} K\|_{op}^q \|f\|_{L^{p,q}}^q + \|f\|_{L^{p,q}}^q \right), \end{aligned}$$

where $D(p, q, \eta, d)^q = \int_{y_k - \frac{\eta}{2}}^{y_k + \frac{\eta}{2}} \left(\int_{E_y} dx \right)^{\frac{q}{p}} dy$. Thus, we have

$$D(p, q, \eta, d) \left(\sum_{k \in \mathbb{Z}} \left(\sum_{j \in \mathbb{Z}^d} |f(x_j, y_k)|^p \right)^{\frac{q}{p}} \right)^{\frac{1}{q}} \leq 2^{1-\frac{1}{q}} \left(1 + \|\operatorname{OSC}_{\frac{\eta}{2}} K\|_{op}^q \right)^{\frac{1}{q}} \|f\|_{L^{p,q}},$$

which shows that

$$\left(\sum_{k \in \mathbb{Z}} \left(\sum_{j \in \mathbb{Z}^d} |f(x_j, y_k)|^p \right)^{\frac{q}{p}} \right)^{\frac{1}{q}} \leq \frac{2^{1-\frac{1}{q}}}{D(p, q, \eta, d)} \left(1 + \|\text{OSC}_{\frac{\eta}{2}} K\|_{op}^q \right)^{\frac{1}{q}} \|f\|_{L^{p,q}}. \quad (4.4)$$

Thus, (4.3) and (4.4) together completes the proof. $\square$

Remark 4.1 We can observe that

$$\sup_k \|\sup_j \|\beta_{j,k}\|_{L_x^1}\|_{L_y^1} \leq \text{Vol}_{d+1}(\gamma),$$

where $\text{Vol}_{d+1}(\gamma)$ is the volume of ball $B_\gamma(x_j, y_k)$ in $\mathbb{R}^{d+1}$. Also

$$\sup_y \left[\sum_{k \in \mathbb{Z}} \sup_j \|\beta_{j,k}(\cdot, y)\|_{L_x^1} \right] \leq \left(\lfloor \frac{2\gamma}{\eta} \rfloor + 1 \right)^{d+1} \text{Vol}_d(\gamma).$$

Thus, one choice of $C(p, q, \gamma, \eta, d)$ is $\left[\left(\lfloor \frac{2\gamma}{\eta} \rfloor + 1 \right)^{d+1} \text{Vol}_d(\gamma) \right]^{\frac{q-p}{pq}} \text{Vol}_{d+1}(\gamma)^{\frac{1}{q}}$. The constant $D(p, q, \eta, d)$ is given by

$$D(p, q, \eta, d) = \left(\frac{\eta}{2} \right)^{\frac{1}{q}} [\text{Vol}_d(\frac{\eta}{2})]^{\frac{1}{p}} B\left(\frac{1}{2}, 1 + \frac{dq}{2p}\right)^{\frac{1}{q}}.$$

Here $B(m, n)$ is the standard Beta function defined by $B(m, n) = 2 \int_0^{\frac{\pi}{2}} (\sin \theta)^{2m-1} (\cos \theta)^{2n-1} d\theta$.

In the following examples, we give a class of kernels satisfying the assumptions of our results.

Example 4.1 Let $\varphi \in W(L^1)$ be continuous and $\tilde{\varphi} \in W(L^1)$ be the dual of φ . Consider the integral kernel K_φ defined by

$$K_\varphi(x, y, r, s) = \sum_{k_2 \in \mathbb{Z}} \sum_{k_1 \in \mathbb{Z}^d} \varphi(x - k_1, y - k_2) \overline{\tilde{\varphi}(r - k_1, s - k_2)}.$$

By simple computation, we obtain

$$\begin{aligned} \sup_y \sup_x \|K_\varphi(x, y, \cdot, \cdot)\|_{L_{r,s}^{1,1}} &\leq \|\varphi\|_{W(L^1)} \|\tilde{\varphi}\|_{L^1} < \infty, \\ \sup_y \sup_r \|K_\varphi(\cdot, y, r, \cdot)\|_{L_x^1} \|_{L_s^1} &\leq \|\varphi\|_{W(L^1)} \|\tilde{\varphi}\|_{W(L^1)} < \infty, \end{aligned}$$

and

$$\sup_s \|\sup_r \|K_\varphi(\cdot, \cdot, r, s)\|_{L_x^1}\|_{L_y^1} \leq \|\varphi\|_{L^1} \|\tilde{\varphi}\|_{W(L^1)} < \infty.$$

Hence, $\|K_\varphi\|_{op}$ is finite. Consider

$$\text{OSC}_\gamma K_\varphi(x, y, r, s) \leq \sum_{k_2 \in \mathbb{Z}} \sum_{k_1 \in \mathbb{Z}^d} \text{osc}_\gamma \varphi(x - k_1, y - k_2) |\tilde{\varphi}(r - k_1, s - k_2)|.$$

From Lemma 2.1, we have $\text{osc}_\gamma \varphi \in W(C, L^1)$, and hence similar to K_φ , we can easily prove that

$$\|\text{OSC}_\gamma K_\varphi\|_{op} < \infty.$$

Thus, for $\|\text{OSC}_\gamma K_\varphi\|_{op} \rightarrow 0$, when $\gamma \rightarrow 0$, it is enough to prove that

$$\sup_y \sup_x \|\text{OSC}_\gamma K_\varphi(x, y, \cdot, \cdot)\|_{L_{r,s}^{1,1}} \rightarrow 0, \text{ when } \gamma \rightarrow 0.$$

Consider

$$\sup_y \sup_x \|\text{OSC}_\gamma K_\varphi(x, y, \cdot, \cdot)\|_{L_{r,s}^{1,1}} \leq \sup_y \sup_x \sum_{k_2 \in \mathbb{Z}} \sum_{k_1 \in \mathbb{Z}^d} \text{osc}_\gamma \varphi(x - k_1, y - k_2) \|\tilde{\varphi}\|_{L^{1,1}}.$$

Now we consider

$$\begin{aligned} & \sup_y \sup_x \sum_{k_2 \in \mathbb{Z}} \sum_{k_1 \in \mathbb{Z}^d} \text{osc}_\gamma \varphi(x - k_1, y - k_2) \\ & \leq \sum_{k_2 \in \mathbb{Z}} \sup_{y \in [0,1]} \sum_{k_1 \in \mathbb{Z}^d} \sup_{x \in [0,1]^d} \text{osc}_\gamma \varphi(x - k_1, y - k_2) \\ & \leq \sum_{k_2 \in \mathbb{Z}} \sum_{k_1 \in \mathbb{Z}^d} \sup_{y \in [0,1]} \sup_{x \in [0,1]^d} \text{osc}_\gamma \varphi(x - k_1, y - k_2) \\ & = \|\text{osc}_\gamma \varphi\|_{W(L^1)}. \end{aligned}$$

Form Lemma 2.1, $\text{osc}_\gamma \varphi \in W(C, L^1)$. We have

$$\sup_y \sup_x \sum_{k_2 \in \mathbb{Z}} \sum_{k_1 \in \mathbb{Z}^d} \text{osc}_\gamma \varphi(x - k_1, y - k_2) < \infty.$$

Thus, for given $\epsilon > 0$, $\exists N_0 > 0$ such that

$$\sup_y \sup_x \sum_{|k_2| \geq N_0} \sum_{|k_1| \geq N_0} \text{osc}_\gamma \varphi(x - k_1, y - k_2) < \frac{\epsilon}{2}.$$

Since φ is continuous, $\exists \delta_0 > 0$ such that for all $|k_1| < N_0, |k_2| < N_0$, we have

$$|\varphi(x + x' - k_1, y + y' - k_2) - \varphi(x - k_1, y - k_2)| \\ < \frac{\epsilon}{2(2N + 1)^{d+1}}, \text{ whenever } (x', y') \in B_{\delta_0}(0, 0).$$

Hence, for $\gamma < \delta_0$, we have

$$\sup_y \sup_x \sum_{|k_2| < N_0} \sum_{|k_1| < N_0} \text{osc}_\gamma \varphi(x - k_1, y - k_2) \leq \frac{\epsilon}{2}.$$

Thus, we obtain

$$\sup_y \sup_x \sum_{k_2 \in \mathbb{Z}} \sum_{k_1 \in \mathbb{Z}^d} \text{osc}_\gamma \varphi(x - k_1, y - k_2) < \epsilon, \quad \forall 0 < \gamma < \delta_0,$$

which implies

$$\lim_{\gamma \rightarrow 0} \|\text{OSC}_\gamma K_\varphi\|_{op} = 0.$$

Example 4.2 Let $\varphi(x, y) = \max \left\{ \frac{1}{2} - |x| - |y|, 0 \right\}$. Consider the sequence of kernels defined by

$$K_N(x, y, r, s) = 4N \sqrt{\frac{3N}{2N-1}} \sum_{k_2 \in \mathbb{Z}} \sum_{k_1 \in \mathbb{Z}} \varphi(x - k_1, y k_2) \\ \times \varphi(N(r - k_1), N(s k_2)), \quad N \geq 1.$$

Then, for $0 < \gamma < \frac{1}{2}$, we have

$$\sup_y \sup_x \|\text{OSC}_\gamma K_N(x, y, \cdot, \cdot)\|_{L_{r,s}^{1,1}} < \frac{12\gamma N^2}{6N^2(2N-1)}, \\ \sup_y \|\sup_r \|\text{OSC}_\gamma K_N(\cdot, y, r, \cdot)\|_{L_x^1}\|_{L_s^1} < \frac{12\gamma N^2}{2N-1},$$

and

$$\sup_s \|\sup_r \|\text{OSC}_\gamma K_N(\cdot, \cdot, r, s)\|_{L_x^1}\|_{L_y^1} < \frac{24\gamma N^2}{2N-1}.$$

Thus, we obtain

$$\|\text{OSC}_\gamma K_N\|_{op} < \frac{12\gamma N^2}{2N-1} \times \left(\frac{1}{6N^2} \right)^{1-\frac{1}{p}} \times 2^{\frac{1}{q}}.$$

Therefore, for $0 < \gamma \leq \frac{1}{12N} - \frac{1}{24N^2}$, we have

$$\|\text{OSC}_\gamma K_N\|_{op} < 1.$$

Example 4.3 Consider $\varphi \in C_c(\mathbb{R}^d \times \mathbb{R})$ with $\|\varphi\|_{L^2} = 1$ such that $\text{supp}(\varphi) \subset \overline{B_{\frac{1}{4}}(0, 0)}$. Let $\psi = \varphi \chi_{\overline{B_{\frac{1}{2}}(0, 0)}}$. Since φ is uniformly continuous, $\sup_{(x, y) \in \overline{B_{\frac{1}{4}}(0, 0)}} \varphi(x, y) = \varphi(a, b)$ for some $(a, b) \in \overline{B_{\frac{1}{4}}(0, 0)}$. Since φ is uniformly continuous, ψ is uniformly continuous. Thus, for $0 < \epsilon < \frac{1}{|\varphi(a, b)|}$, $\exists \delta_0 > 0$ such that for every $(x, y) \in \overline{B_{\frac{1}{2}}(0, 0)}$,

$$|\psi(x', y') - \psi(x, y)| < \epsilon, \text{ whenever } (x', y') \in B_{\delta_0}(x, y).$$

Let $K(x, y, r, s) = \sum_{k_2 \in \mathbb{Z}} \sum_{k_1 \in \mathbb{Z}^d} \varphi(x - k_1, y - k_2) \overline{\varphi(r - k_1, s - k_2)}$. If we choose $0 < \gamma < \min\{\frac{1}{4}, \delta_0\}$, then

$$\begin{aligned} \sup_y \sup_x \|\text{OSC}_\gamma K(x, y, \cdot, \cdot)\|_{L_{r,s}^{1,1}} &< 1, \\ \sup_y \|\sup_r \|\text{OSC}_\gamma K(\cdot, y, r, \cdot)\|_{L_x^1}\|_{L_s^1} &< 1, \end{aligned}$$

and

$$\sup_s \|\sup_r \|\text{OSC}_\gamma K(\cdot, \cdot, r, s)\|_{L_x^1}\|_{L_y^1} < 1.$$

Thus, we obtain

$$\|\text{OSC}_\gamma K\|_{op} < 1, \text{ for every } 0 < \gamma < \min\{\frac{1}{4}, \delta_0\}.$$

In Theorem 4.1, we have established a sampling inequality for functions belonging in V . Our next goal is to reconstruct a function $f \in V$ from its samples $\{f(x_j, y_k) : j \in \mathbb{Z}^d, k \in \mathbb{Z}\}$. In the following theorem, we give an iterative approximation-projection based algorithm for reconstruction of $f \in V$ from its samples. First, we approximate the function f from its samples by using quasi approximation Qf and then project it to the space V to get the initial approximation f_0 . Now we apply the approximation-projection algorithm on the error $e = f - f_0$ to get the correction e_0 and update the first approximation as $f_1 = f_0 + e_0$. By repeating this process we obtain a sequence f_n which converges to f exponentially. In order to get such convergence, we need to choose sampling set to be dense enough which is given in the following theorem.

Theorem 4.2 Let $1 \leq p \leq q < \infty$ and K be an integral kernel satisfying (3.2)–(3.4). If P is a bounded projection from $L^{p,q}$ onto V and $X = \{(x_j, y_k) : j \in \mathbb{Z}^d, k \in \mathbb{Z}\}$ is a separated γ -dense set such that

$$\|P\|_{L^{p,q}} \|\text{OSC}_\gamma K\|_{op} < 1,$$

then every $f \in V$ can be uniquely reconstructed from its samples $\{f(x_j, y_k) : j \in \mathbb{Z}^d, k \in \mathbb{Z}\}$ by using the following iterative reconstruction algorithm

$$\begin{aligned} f_0 &= PQf, \\ f_{n+1} &= f_n + PQ(f - f_n), \quad n \geq 0. \end{aligned}$$

Then $f_n \rightarrow f$ and the error estimate after n^{th} iteration is given by

$$\|f - f_n\|_{L^{p,q}} \leq (\|P\|_{L^{p,q}} \|OSC_\gamma K\|_{op})^{n+1} \|f\|_{L^{p,q}}.$$

Proof For any $f \in V$, we have

$$\begin{aligned} \|f - PQf\|_{L^{p,q}} &= \|Pf - PQf\|_{L^{p,q}} \\ &\leq \|P\|_{L^{p,q}} \|f - Qf\|_{L^{p,q}} \\ &\leq \|P\|_{L^{p,q}} \|OSC_\gamma K\|_{op} \|f\|_{L^{p,q}}, \end{aligned}$$

where last inequality is due to (4.1). Hence the result follows from Proposition 2.3. $\square$

5 Average sampling and reconstruction in V

In this section, we study the average sampling problem in the function space V . In the following theorem, we establish the average sampling inequality for $f \in V$. In particular, we extend the Theorem 4.1 for the case of average sampling. To get the average sampling inequality, we first approximate the function f from its average samples by using the quasi approximation Af . Then we estimate the $\|f - Af\|_{L^{p,q}}$ and $\|Af\|_{L^{p,q}}$ to get the required lower bound in the average sampling inequality. We prove the following result.

Theorem 5.1 *Let $1 \leq p \leq q < \infty$ and $X = \{(x_j, y_k) : j \in \mathbb{Z}^d, k \in \mathbb{Z}\}$ be a separated γ -dense set and $\{\psi_{x_j, y_k} : j \in \mathbb{Z}^d, k \in \mathbb{Z}\}$ be a collection of averaging functions such that $\text{supp } \psi_{x_j, y_k} \subset B_\delta(x_j, y_k)$. If K is an integral kernel satisfying (3.2)–(3.4), γ and δ are positive real numbers such that $M\|OSC_{\gamma+\delta} K\|_{op} < 1$, then we have the following average sampling inequality*

$$\begin{aligned} \frac{(1 - M\|OSC_{\gamma+\delta} K\|_{op})}{C(p, q, \gamma, \eta, d)} \|f\|_{L^{p,q}} &\leq \left(\sum_{k \in \mathbb{Z}} \left(\sum_{j \in \mathbb{Z}^d} |\langle f, \psi_{x_j, y_k} \rangle|^p \right)^{\frac{q}{p}} \right)^{\frac{1}{q}} \\ &\leq \frac{2^{1-\frac{1}{q}} (1 + \|M^q OSC_{\frac{\gamma}{2}+\delta} K\|_{op}^q)^{\frac{1}{q}}}{D(p, q, \eta, d)} \|f\|_{L^{p,q}}, \end{aligned}$$

for every $f \in V$.

Proof For any $f \in V$, define the approximation operator

$$Af = \sum_{k \in \mathbb{Z}} \sum_{j \in \mathbb{Z}^d} \langle f, \psi_{x_j, y_k} \rangle \beta_{j,k}.$$

Since

$$\begin{aligned} |f(x, y) - Af(x, y)| &\leq \sum_{k \in \mathbb{Z}} \sum_{j \in \mathbb{Z}^d} |f(x, y) - \langle f, \psi_{x_j, y_k} \rangle \beta_{j,k}(x, y)| \\ &\leq \sum_{k \in \mathbb{Z}} \sum_{j \in \mathbb{Z}^d} \int_{\mathbb{R}^{d+1}} |f(x, y) - f(t)| |\psi_{x_j, y_k}(t)| dt \beta_{j,k}(x, y) \\ &\leq \sum_{k \in \mathbb{Z}} \sum_{j \in \mathbb{Z}^d} \text{osc}_{\gamma+\delta} f(x, y) \int_{\mathbb{R}^{d+1}} |\psi_{x_j, y_k}(t)| dt \beta_{j,k}(x, y) \\ &\leq M \sum_{k \in \mathbb{Z}} \sum_{j \in \mathbb{Z}^d} \text{osc}_{\gamma+\delta} f(x, y) \beta_{j,k}(x, y) \\ &= M \text{osc}_{\gamma+\delta} f(x, y). \end{aligned}$$

We have

$$\|f - Af\|_{L^{p,q}}^q \leq M^q \|\text{osc}_{\gamma+\delta} f\|_{L^{p,q}}^q.$$

Since

$$\text{osc}_{\gamma+\delta} f(x, y) \leq \int_{\mathbb{R}} \int_{\mathbb{R}^d} \text{OSC}_{\gamma+\delta} K(x, y, r, s) |f(r, s)| dr ds,$$

using the similar idea as in Theorem 3.1, we obtain

$$\begin{aligned} &\|f - Af\|_{L^{p,q}} \\ &\leq \sup_y \sup_x \|\text{OSC}_{\gamma+\delta} K(x, y, \cdot, \cdot)\|_{L_{r,s}^{1,1}}^{\frac{1}{p'}} \times \sup_y \|\sup_r \|\text{OSC}_{\gamma+\delta} K(\cdot, y, r, \cdot)\|_{L_x^1}\|_{L_s^1}^{\frac{q-p}{pq}} \\ &\quad \times \sup_s \|\sup_r \|\text{OSC}_{\gamma+\delta} K(\cdot, \cdot, r, s)\|_{L_x^1}\|_{L_y^1}^{\frac{1}{q}} M \|f\|_{L^{p,q}} \\ &= M \|\text{OSC}_{\gamma+\delta} K\|_{op} \|f\|_{L^{p,q}}. \end{aligned} \quad (5.1)$$

By using the similar idea as in Theorem 4.1, we obtain

$$\|Af\|_{L^{p,q}} \leq C(p, q, \gamma, \eta, d) \left(\sum_{k \in \mathbb{Z}} \left(\sum_{j \in \mathbb{Z}^d} |\langle f, \psi_{x_j, y_k} \rangle|^p \right)^{\frac{q}{p}} \right)^{\frac{1}{q}}, \quad (5.2)$$

where $C(p, q, \gamma, \eta, d) = \sup_y \left[\sum_{k \in \mathbb{Z}} \sup_j \|\beta_{j,k}(\cdot, y)\|_{L_x^1} \right]^{\frac{q-p}{pq}} \sup_k \|\sup_j \|\beta_{j,k}\|_{L_x^1}\|_{L_y^1}^{\frac{1}{q}}$.

Using (5.1) and (5.2), we have

$$\left(\sum_{k \in \mathbb{Z}} \left(\sum_{j \in \mathbb{Z}} |\langle f, \psi_{x_j, y_k} \rangle|^p \right)^{\frac{q}{p}} \right)^{\frac{1}{q}} \geq \left(\frac{1 - M \|\text{OSC}_{\gamma+\delta} K\|_{op}}{C(p, q, \gamma, \eta, d)} \right) \|f\|_{L^{p,q}}, \quad (5.3)$$

for every $f \in V$.

For the other side inequality, we consider

$$\begin{aligned} \int_{E_y} |\langle f, \psi_{x_j, y_k} \rangle|^p dx &\leq 2^{p-1} \left[\int_{E_y} |\langle f, \psi_{x_j, y_k} \rangle - f(x, y)|^p dx + \int_{E_y} |f(x, y)|^p dx \right] \\ &\leq 2^{p-1} \left[\int_{E_y} |M \text{osc}_{\frac{\eta}{2}+\delta} f(x, y)|^p dx + \int_{E_y} |f(x, y)|^p dx \right], \end{aligned}$$

for $(x, y) \in B_{\frac{\eta}{2}}(x_j, y_k)$. By taking summation over $j \in \mathbb{Z}^d$, we get

$$\sum_{j \in \mathbb{Z}^d} |\langle f, \psi_{x_j, y_k} \rangle|^p \int_{E_y} dx \leq 2^{p-1} \left[\int_{\mathbb{R}^d} |M \text{osc}_{\frac{\eta}{2}+\delta} f(x, y)|^p dx + \int_{\mathbb{R}^d} |f(x, y)|^p dx \right].$$

Now following the similar path as in Theorem 4.1, we can prove that

$$\begin{aligned} &\left(\sum_{k \in \mathbb{Z}} \left(\sum_{j \in \mathbb{Z}} |\langle f, \psi_{x_j, y_k} \rangle|^p \right)^{\frac{q}{p}} \right)^{\frac{1}{q}} \\ &\leq \frac{2^{1-\frac{1}{q}}}{D(p, q, \eta, d)} \left(1 + M^q \|\text{OSC}_{\frac{\eta}{2}+\delta} K\|_{op}^q \right)^{\frac{1}{q}} \|f\|_{L^{p,q}}. \end{aligned} \quad (5.4)$$

Thus, (5.3) and (5.4) together completes the proof. $\square$

In the above Theorem 5.1, we have established an average sampling inequality for $f \in V$. Similar to Theorem 4.2, a reconstruction algorithm for the recovery any $f \in V$ from its average samples is given in the following theorem with exponential convergence.

Theorem 5.2 *Let $1 \leq p \leq q < \infty$ and K be an integral kernel satisfying (3.2)–(3.4). If $X = \{(x_j, y_k) : j \in \mathbb{Z}^d, k \in \mathbb{Z}\}$ is a separated γ -dense set and $\{\psi_{x_j, y_k} : j \in \mathbb{Z}^d, k \in \mathbb{Z}\}$ is a collection of averaging functions such that $\text{supp } \psi_{x_j, y_k} \subset B_\delta(x_j, y_k)$. If P is a bounded projection from $L^{p,q}$ onto V , γ and δ are positive real numbers such that $M \|P\|_{L^{p,q}} \|\text{OSC}_{\gamma+\delta} K\|_{op} < 1$, then every $f \in V$ can be uniquely reconstructed*

from its average samples $\{f, \psi_{x_j, y_k}\} : j \in \mathbb{Z}^d, k \in \mathbb{Z}\}$ by using the following iterative reconstruction algorithm

$$\begin{aligned} f_0 &= PAf, \\ f_{n+1} &= f_n + PA(f - f_n), \quad n \geq 0. \end{aligned}$$

Then $f_n \rightarrow f$ and the error estimate after n^{th} iteration is given by

$$\|f - f_n\|_{L^{p,q}} \leq (\|P\|_{L^{p,q}} \|\text{OSC}_{\gamma+\delta} K\|_{op})^{n+1} \|f\|_{L^{p,q}}.$$

Proof For any $f \in V$, we have

$$\begin{aligned} \|f - PAf\|_{L^{p,q}} &= \|Pf - PAf\|_{L^{p,q}} \\ &\leq \|P\|_{L^{p,q}} \|f - Af\|_{L^{p,q}} \\ &\leq M \|P\|_{L^{p,q}} \|\text{OSC}_{\gamma+\delta} K\|_{op} \|f\|_{L^{p,q}}, \end{aligned}$$

where last inequality is due to (5.1). Hence the result follows from Proposition 2.3. $\square$

6 Error estimates of iterative reconstruction algorithm

In this section, we first consider the error estimate when sample values are corrupted with the noise, i.e., the sample values of $f \in V$ are taken as $\{f(x_j, y_k) + \epsilon(x_j, y_k) : j \in \mathbb{Z}^d, k \in \mathbb{Z}\}$. Then, we consider the error estimate when iterative reconstruction algorithm is corrupted, i.e., each iteration is corrupted by some noise. In particular, we obtain the following Theorems 6.1 and 6.2; The proof follows in same manner as in [20, Theorem 4.1 and Theorem 4.2].

Theorem 6.1 Let $1 \leq p \leq q < \infty$ and K be an integral kernel satisfying (3.2)–(3.4). If P is a bounded projection from $L^{p,q}$ onto V and $X = \{(x_j, y_k) : j \in \mathbb{Z}^d, k \in \mathbb{Z}\}$ is a separated γ -dense set such that

$$\|P\|_{L^{p,q}} \|\text{OSC}_{\gamma} K\|_{op} < 1,$$

then

$$\|f - f'\|_{L^{p,q}} \leq \frac{C(p, q, \gamma, \eta, d) \|P\|_{L^{p,q}}}{1 - \|P\|_{L^{p,q}} \|\text{OSC}_{\gamma} K\|_{op}} \|\epsilon\|_{\ell^{p,q}},$$

where $C(p, q, \gamma, \eta, d)$ is given in (4.2), $\epsilon = \{\epsilon(x_j, y_k) : j \in \mathbb{Z}^d, k \in \mathbb{Z}\} \in \ell^{p,q}$, and f' is a reconstructed signal from the corrupted samples $\{f(x_j, y_k) + \epsilon(x_j, y_k) : j \in \mathbb{Z}^d, k \in \mathbb{Z}\}$ by using the iterative reconstruction algorithm given in Theorem 4.2.

Theorem 6.2 Let $1 \leq p \leq q < \infty$ and K be an integral kernel satisfying (3.2)–(3.4). Assume that P is a bounded projection from $L^{p,q}$ onto V and $X = \{(x_j, y_k) : j \in \mathbb{Z}^d, k \in \mathbb{Z}\}$ is a separated γ -dense set such that

$$\|P\|_{L^{p,q}} \|OSC_\gamma K\|_{op} < 1,$$

and the iterative reconstruction algorithm for the reconstruction of $f \in V$,

$$\begin{aligned}\tilde{f}_0 &= PQf + \epsilon_0, \\ \tilde{f}_{n+1} &= \tilde{f}_0 + (I - PQ)\tilde{f}_n + \epsilon_{n+1}, \quad n \geq 0,\end{aligned}$$

is corrupted by the noise $\epsilon_n \in V$, $n \geq 0$. Then

$$\begin{aligned}\|f - \tilde{f}_n\|_{L^{p,q}} &\leq \left(1 + \frac{C(p, q, \gamma, \eta, d) \|P\|_{L^{p,q}}}{1 - \|P\|_{L^{p,q}} \|OSC_\gamma K\|_{op}}\right) \left(n \|\epsilon_0\|_{L^{p,q}} + \sum_{j=0}^n \|\epsilon_j\|_{L^{p,q}}\right) \\ &\quad + \left(\|P\|_{L^{p,q}} \|OSC_\gamma K\|_{op}\right)^{n+1} \|f\|_{L^{p,q}},\end{aligned}$$

where $C(p, q, \gamma, \eta, d)$ is given in (4.2).

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References

1. Aldroubi, A., Gröchenig, K.: Beurling–Landau-type theorems for nonuniform sampling in shift-invariant spline spaces. *J. Fourier Anal. Appl.* **6**(1), 93–103 (2000)
2. Aldroubi, A., Gröchenig, K.: Nonuniform sampling and reconstruction in shift-invariant spaces. *SIAM Rev.* **43**(4), 585–620 (2001)
3. Aldroubi, A., Sun, Q., Tang, W.S.: p -frames and shift invariant subspaces of L^p . *J. Fourier Anal. Appl.* **7**(1), 1–22 (2001)
4. Aldroubi, A., Sun, Q., Tang, W.S.: Convolution, average sampling, and a Calderon resolution of the identity for shift-invariant spaces. *J. Fourier Anal. Appl.* **11**(2), 215–244 (2005)
5. Benedek, A., Panzone, R.: The space L^p with mixed norm. *Duke Math. J.* **28**(3), 301–324 (1961)
6. Bhandari, A., Zayed, A.I.: Shift-invariant and sampling spaces associated with the fractional Fourier transform domain. *IEEE Trans. Signal Process.* **60**(4), 1627–1637 (2012)
7. Bhandari, A., Zayed, A.I.: Shift-invariant and sampling spaces associated with the special affine Fourier transform. *Appl. Comput. Harmon. Anal.* (2017). <https://doi.org/10.1016/j.acha.2017.07.002>
8. Butzer, P.L., Stens, R.L.: Sampling theory for not necessarily band-limited functions: a historical overview. *SIAM Rev.* **34**(1), 40–53 (1992)
9. Cheng, C., Jiang, Y., Sun, Q.: Spatially distributed sampling and reconstruction. *Appl. Comput. Harmon. Anal.* **47**, 109–148 (2019)
10. Fernandez, D.L.: Vector-valued singular integral operators on L^p -spaces with mixed norms and applications. *Pac. J. Math.* **129**(2), 257–275 (1987)
11. Francia, J.L., Ruiz, F.J., Torrea, J.L.: Calderón–Zygmund theory for operator-valued kernels. *Adv. Math.* **62**(1), 7–48 (1986)
12. Gröchenig, K., Romero, J.L., Stöckler, J.: Sampling theorems for shift-invariant spaces, Gabor frames, and totally positive functions. *Invent. math.* **211**(3), 1119–1148 (2018)
13. Gröchenig, K., Stöckler, J.: Gabor frames and totally positive functions. *Duke Math. J.* **162**(6), 1003–1031 (2013)
14. Jiang, Y., Sun, W.: Adaptive sampling of time-space signals in a reproducing kernel subspace of mixed lebesgue space. *arXiv preprint arXiv:1904.00727* (2019)

15. Kulkarni, S.H., Radha, R., Sivananthan, S.: Non-uniform sampling problem. *J. Appl. Funct. Anal.* **4**(1), 58–74 (2009)
16. Kumar, A., Sampath, S.: Average sampling and reconstruction in shift-invariant spaces and variable bandwidth spaces. *Appl. Anal.* (2018). <https://doi.org/10.1080/00036811.2018.1508652>
17. Li, R., Liu, B., Liu, R., Zhang, Q.: Nonuniform sampling in principal shift-invariant subspaces of mixed Lebesgue spaces $L^{p,q}(\mathbb{R}^{d+1})$. *J. Math. Anal. Appl.* **453**(2), 928–941 (2017)
18. Moumni, T., Zayed, A.I.: A generalization of the prolate spheroidal wave functions with applications to sampling. *Integral Transforms Spec. Funct.* **25**(6), 433–447 (2014)
19. Nashed, M.Z., Sun, Q.: Sampling and reconstruction of signals in a reproducing kernel subspace of $L^p(\mathbb{R}^d)$. *J. Funct. Anal.* **258**(7), 2422–2452 (2010)
20. Nashed, M.Z., Sun, Q., Xian, J.: Convolution sampling and reconstruction of signals in a reproducing kernel subspace. *Proc. Am. Math. Soc.* **141**(6), 1995–2007 (2013)
21. Razafinjatovo, H.N.: Iterative reconstructions in irregular sampling with derivatives. *J. Fourier Anal. Appl.* **1**(3), 281–295 (1994)
22. Samarah, S., Obeidat, S., Salman, R.: A schur test for weighted mixed-norm spaces. *Anal. Math.* **31**(4), 277–289 (2005)
23. Selvan, A.A., Radha, R.: Sampling and reconstruction in shift-invariant spaces of B -spline functions. *Acta Appl. Math.* **145**(1), 175–192 (2016)
24. Selvan, A.A., Radha, R.: An optimal result for sampling density in shift-invariant spaces generated by Meyer scaling function. *J. Math. Anal. Appl.* **451**(1), 197–208 (2017)
25. Sun, Q.: Nonuniform average sampling and reconstruction of signals with finite rate of innovation. *SIAM J. Math. Anal.* **38**(5), 1389–1422 (2006)
26. Sun, W., Zhou, X.: Reconstruction of band-limited functions from local averages. *Constr. Approx.* **18**(2), 205–222 (2002)
27. Sun, W., Zhou, X.: Reconstruction of functions in spline subspaces from local averages. *Proc. Am. Math. Soc.* **131**(8), 2561–2571 (2003)
28. Tao, R., Li, B.Z., Wang, Y., Aggrey, G.K.: On sampling of band-limited signals associated with the linear canonical transform. *IEEE Trans. Signal Process.* **56**(11), 5454–5464 (2008)
29. Vetterli, M., Marziliano, P., Blu, T.: Sampling signals with finite rate of innovation. *IEEE Trans. Signal Process.* **50**(6), 1417–1428 (2002)
30. Walter, G.: A sampling theorem for wavelet subspaces. *IEEE Trans. Inf. Theory* **38**(2), 881–884 (1992)
31. Walter, G., Shen, X.: Sampling with prolate spheroidal wave functions. *Sampl. Theory Signal Image Process.* **2**(1), 25–52 (2003)
32. Ward, E.L.: New estimates in harmonic analysis for mixed Lebesgue spaces. Ph.D. thesis, University of Kansas (2010)
33. Wei, D., Ran, Q., Li, Y.: Generalized sampling expansion for band-limited signals associated with the fractional Fourier transform. *IEEE Signal Process. Lett.* **17**(6), 595–598 (2010)
34. Xian, J.: Weighted sampling and reconstruction in weighted reproducing kernel spaces. *J. Math. Anal. Appl.* **367**(1), 34–42 (2010)
35. Xian, J., Luo, S.P., Lin, W.: Weighted sampling and signal reconstruction in spline subspaces. *Signal Process.* **86**(2), 331–340 (2006)
36. Zayed, A.I.: Sampling of signals band-limited to a disc in the linear canonical transform domain. *IEEE Signal Process. Lett.* **25**(12), 1765–1769 (2018)
37. Zhang, Q.: Nonuniform average sampling in multiply generated shift-invariant subspaces of mixed lebesgue spaces. arXiv preprint [arXiv:1806.05055](https://arxiv.org/abs/1806.05055) (2018)